Identifiability of B's row from endpoint counts: what can be proved, and what the E0 experiments actually show

BIO-SBI — note accompanying `report_sequential`, Part I

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Abstract

The sequential report (Part I: E0, X5a, X5c, E0 at 4096 clones) concludes that, with B divisions free, B's row is *structurally* non-identifiable from endpoint counts. We study the Galton–Watson (GW) model formally and with the exact likelihood (probability generating functions, no simulation). The results split the claim in two. (i) *What is true and provable*: B's row enters the mean dynamics only through its mean offspring vector, so first moments are blind to a 3-dimensional family that trades switching for division (Theorem 2); within that family the row is fixed by its second factorial moments, and B divisions always inflate clone-size variance (Theorem 3); and the variance of the Kesten–Stigum limit sees the second moments of *both* rows through a single scalar, so A's row can absorb B's (Theorem 4). (ii) *What is not true*: the model is not structurally non-identifiable. The exact Fisher information at the truth is positive definite at every endpoint $n = 2, \ldots, 12$ (Computation 6), so the parameters are locally identifiable. The non-identifiability is *practical*. At the real-like design the Cramér–Rao bound on B→A is 0.28, larger than the prior sd. Globally, the best model in which B divides with probability 0.5 is only 0.36 nats worse than the truth over all 266 organoids (Computation 9). At 4096 clones on three endpoints the bound is 0.023, and the exact likelihood rejects the NPE answer by 116–178 nats on the very datasets used in the report; the `moments` summaries still separate it (97% accuracy). That failure is in the amortised inference, not in the model; exact posterior sampling on the same data returns B→A $= 0.78 \pm 0.02$ (5 seeds). Finally, the information about B's row decays geometrically with the endpoint generation. Locally, one endpoint at generation 2–3 gives 10–15 times the Fisher information of the real-like design with the same organoids; globally the likelihood flattens far from the truth, so the gain is 7–18 times in evidence and still not decisive at 266 organoids.

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1 Setting

Types A (= LGR5$^+$, index 1) and B (= KRT20$^+$, index 2). Each living cell has one event per generation, drawn from its type's *row*: death, persist, switch, or division into AA, AB, BB. Write the rows as

$$\pi_A = (d_A, p_A, t_A, a_A, b_A, c_A), \qquad \pi_B = (d_B, p_B, q, a_B, b_B, c_B),$$

in the order death, persist, switch (A→B resp. B→A), div_AA, div_AB, div_BB. With $s = (x, y)$ marking A and B, the offspring probability generating functions (pgfs) are

$$f_A(x, y) = d_A + p_A x + t_A y + a_A x^2 + b_A xy + c_A y^2,$$
$$f_B(x, y) = d_B + p_B y + q\, x + a_B x^2 + b_B xy + c_B y^2.$$

$Z_n = (Z_{n,A}, Z_{n,B})$ is the clone at generation n from one A founder; its pgf is F_n^A with $F_n = f \circ F_{n-1}$, $F_0 = \mathrm{id}$. An endpoint n contributes N_n i.i.d. clones from the law of Z_n *conditioned on* $Z_n \neq 0$ (detected clones, perfect capture). A design is $\mathcal{D} = \{(n, N_n)\}$. The free parameter is $\theta = (\pi_A, \pi_B) \in \Delta_5 \times \Delta_5$ (dimension 10; `two_type`). Fixing $a_B = b_B = c_B = 0$ gives `reversible_ab` (dimension 7).

The truth used throughout E0/X5/E4 is

$$\theta^\star : \ \pi_A = (.03, .03, .05, .55, .05, .29), \quad \pi_B = (.10, .10, .80, 0, 0, 0),$$

which lies on the boundary of Δ_5 for B (no B divisions). Mean matrix $M_{kj} = \partial_j f_k(1, 1)$, Perron root ρ, right/left Perron vectors r, u ($Mr = \rho r$, $uM = \rho u$, $u \cdot r = 1$). At $\theta^\star$: $M = \left(\begin{smallmatrix} 1.18 & 0.68 \\ 0.80 & 0.10 \end{smallmatrix}\right)$, $\rho = 1.554$, $\lambda_2 = -0.274$, $r \propto (1, 0.550)$.

Three notions of identifiability. We separate them because the report's word "structural" conflates them.

Definition 1. (a) θ is *structurally (globally) identifiable* from design $\mathcal{D}$ if $\theta \mapsto (\mathcal{L}(Z_n \mid Z_n \neq 0))_{n \in \mathcal{D}}$ is injective. (b) It is *locally identifiable* at $\theta^\star$ if this holds in a neighbourhood; a sufficient condition is a positive definite Fisher information $I_{\mathcal{D}}(\theta^\star) = \sum_n N_n I_n(\theta^\star)$. (c) A parameter is *practically non-identifiable* at design $\mathcal{D}$ if it is locally identifiable but the attainable precision, e.g. the Cramér–Rao (CR) bound $\sqrt{g^\top I_{\mathcal{D}}^{-1} g}$ for a functional with gradient g, is no better than the prior spread.

A model that is structurally non-identifiable stays so for every N. A practically non-identifiable one becomes identifiable as N grows, at rate $N^{-1/2}$ (Bernstein–von Mises), and better designs move it.

Background and references. Probability generating functions and their derivatives: [B7, Ch. XI]. The composition rule $F_n = f \circ F_{n-1}$: single type [B1, Ch. I], [B2, Ch. I]; several types [B1, Ch. V], [B2, Ch. II]; for biologists [B3]. Mean matrix and Perron–Frobenius theory for branching processes: [B1, Ch. V], [B2, Ch. II]. Identifiability: non-singular Fisher information implies local identifiability [I1]; structural vs practical [I2, I3]. Bernstein–von Mises and the $N^{-1/2}$ rate: [I6, Ch. 10]. Cramér–Rao bound for a function of the parameters: [I5, Ch. 2], [I6, Ch. 8].

2 Exact results on moments

Theorem 2 (Mean fibres)**.** *For every n, $\mathbb{E}Z_n = e_A M^n$, and B's row enters M only through its mean offspring vector*

$$\mu_B = (M_{BA}, M_{BB}) = (q + 2a_B + b_B,\ p_B + b_B + 2c_B).$$

For fixed μ_B the admissible B rows form a polytope of dimension 3 (when μ_B is interior), spanned locally by the moves

$$
\begin{aligned}
&e_1:\ q \to q-2\varepsilon,\ a_B \to a_B+\varepsilon,\ d_B \to d_B+\varepsilon && \textit{(switch} \to \textit{divide-into-AA and die)},\\
&e_2:\ p_B \to p_B-2\varepsilon,\ c_B \to c_B+\varepsilon,\ d_B \to d_B+\varepsilon && \textit{(persist} \to \textit{divide-into-BB and die)},\\
&e_3:\ q \to q-\varepsilon,\ p_B \to p_B-\varepsilon,\ b_B \to b_B+\varepsilon && \textit{(switch and persist} \to \textit{divide-into-AB)}.
\end{aligned}
$$

Every expected count at every endpoint, and therefore every statistic that depends on the data only through mean counts, is constant on these fibres.

Proof. $\mathbb{E}[Z_n \mid Z_{n-1}] = Z_{n-1}M$ gives $\mathbb{E}Z_n = e_A M^n$. The two entries of μ_B are linear in π_B, and π_B lies in a 5-dimensional simplex. So the fibre is the intersection of the simplex with an affine subspace of codimension 2, which has dimension 3 at interior points. Each e_i keeps $\sum \pi_B = 1$ and both entries of μ_B; they are linearly independent. □

At the truth only the directions with $\varepsilon \geq 0$ are admissible (B divisions ≥ 0). The C2ST alternative of E0 (B→AA $= a$, B→A $= 0.8 - 2a$, B death $= 0.1 + a$) is exactly e_1.

Background and references. $\mathbb{E}Z_n = e_A M^n$ for multitype Galton–Watson processes: [B1, Ch. V], [B2, Ch. II], [B4]. The same "only combinations of fate probabilities are fixed by average clone growth" phenomenon in clonal data: [C2], [C3].

Theorem 3 (Second moments fix the row; B divisions inflate variance)**.** *Let $G_k = \mathbb{E}[\xi_k \xi_k^\top] - \operatorname{diag}(\mu_k)$ be the second factorial moment matrix of type k's offspring ξ_k. Then*

$$
G_k = \begin{pmatrix} 2a_k & b_k \\ b_k & 2c_k \end{pmatrix} \qquad (k = A, B),
$$

and $\pi_k \mapsto (\mu_k, G_k)$ is an affine bijection from Δ_5 onto its image. Moreover, with $V_k = \operatorname{Cov}(\xi_k)$ and $m_j = e_A M^j$,

$$
\operatorname{Cov}(Z_n) = \sum_{j=0}^{n-1} (M^\top)^{n-1-j} \Big(\sum_k m_j(k) V_k \Big) M^{n-1-j}. \tag{1}
$$

Along a mean fibre (Theorem 2) only $V_B = G_B + \operatorname{diag}(\mu_B) - \mu_B \mu_B^\top$ changes, by ΔG_B. Starting from a row without B divisions (as $\theta^\star$), every admissible move has $\Delta G_B = \left(\begin{smallmatrix} 2a_B & b_B \\ b_B & 2c_B \end{smallmatrix}\right)$ with $a_B, b_B, c_B \geq 0$. Hence, for every $w \in \mathbb{R}^2_{>0}$ (e.g. $w = (1,1)$, total clone size) and $n \geq 2$,

$$
\operatorname{Var}(w \cdot Z_n) \text{ strictly increases with the B-division mass } \delta = a_B + b_B + c_B.
$$

Proof. The formula for G_k is a direct computation over the six outcomes: only the divisions have two daughters, and $\mathbb{E}[\xi_A(\xi_A - 1)] = 2a_k$, $\mathbb{E}[\xi_A \xi_B] = b_k$, $\mathbb{E}[\xi_B(\xi_B - 1)] = 2c_k$. Given (μ_k, G_k), the three division probabilities are read off G_k, then persist and switch off μ_k, and death is the remainder; so the map is an affine bijection. Formula (1) is the standard covariance recursion $\operatorname{Cov}(Z_n) = M^\top \operatorname{Cov}(Z_{n-1}) M + \sum_k \mathbb{E}Z_{n-1,k} V_k$ (Harris 1963; Athreya and Ney 1972, Ch. V), unrolled. Along a fibre M and μ_B are fixed, so the change is

$$
\Delta \operatorname{Var}(w \cdot Z_n) = \sum_{j=1}^{n-1} m_j(B)\, v_j^\top \Delta G_B v_j, \qquad v_j = M^{n-1-j} w > 0,
$$

and $v^\top \Delta G_B v = 2a_B v_A^2 + 2b_B v_A v_B + 2c_B v_B^2 > 0$ whenever $v > 0$ and $\delta > 0$. Since $t_A + b_A + c_A > 0$ we have $m_j(B) > 0$ for $j \geq 1$. □

Theorems 2–3 explain why the E0 C2ST found the switching/division difference only in clone-size *variability*. They also explain why `CohortMeans` cannot work. Its 6 features for 10 parameters are rank deficient by construction, and E0 showed that variance carries the information.

Background and references. Factorial moments as derivatives of the pgf at 1: [B7, Ch. XI]. Second-moment (covariance) recursion for multitype processes: [B2, Ch. II], [B1, Ch. V]; a gentler derivation in [B3]. Clone-size variability as the informative quantity in lineage-tracing data: [C2], [C3].

Theorem 4 (The variance of the growth limit sees one scalar)**.** *Let $\rho > 1$ and $W = \lim_n r \cdot Z_n/\rho^n$ (a.s. and in L^2, since the offspring have finite variance). Then*

$$\operatorname{Var} W = \sum_{k\in\{A,B\}} \beta_k \, r^\top V_k r, \qquad \beta_k = \sum_{j\geq 0} \rho^{-2(j+1)} m_j(k).$$

Hence, on the product of the two mean fibres (both rows moved with M fixed), $\operatorname{Var} W$ *depends on the six second factorial moments (G_A, G_B) only through the scalar*

$$\beta_A \, r^\top G_A r + \beta_B \, r^\top G_B r.$$

An increase of B's division variance can be cancelled exactly by fewer A divisions. At $\theta^\star$, $\beta =$ (1.000, 0.294).

Proof. $W_n = r \cdot Z_n/\rho^n$ is a martingale. Its increments $W_{j+1} - W_j = \rho^{-(j+1)} \sum_{\text{cells } c \text{ in gen } j}(r \cdot \xi_c - \rho r_{k(c)})$ are uncorrelated with conditional variance $\rho^{-2(j+1)} \sum_k Z_{j,k} \, r^\top V_k r$. Take expectations and sum (orthogonality of martingale increments). $W_0 = r_A$ is deterministic. Along fibres M, hence ρ, r, m_j and β, are fixed, and $r^\top V_k r = r^\top G_k r + \sum_i \mu_k(i) r_i^2 - (\mu_k \cdot r)^2$ changes only through $r^\top G_k r$. □

Check against the exact law (`scripts/b_row_theory/varW.py`): $\operatorname{Var}(r \cdot Z_n/\rho^n) = 0.3263$ ($n = 10$), 0.3286 ($n = 12$) against the formula $\operatorname{Var} W = 0.3302$.

Background and references. The martingale $r \cdot Z_n/\rho^n$ and its limit W: [B1, Ch. V], [B5]; L^2 convergence and the variance of W when offspring have finite variance: [B2, Ch. II], [B1, Ch. V]. Orthogonality of martingale increments (the step $\operatorname{Var} W = \sum_j \mathbb{E}(\text{increment}_j^2)$): [P1, Ch. 12]. The single-type version of the formula, $\operatorname{Var} W = \sigma^2/(m^2 - m)$, is in [B1, Ch. I] and [B2, Ch. I] and is a good warm-up.

Remark 5 (What Theorem 4 does and does not explain). For large clones $Z_n \approx \rho^n W u$. The dominant second-order feature of late endpoints, the spread of clone sizes, therefore constrains A's and B's division variances only jointly. This is the mechanism by which leaving B divisions free biases A's row (E0 at 4096: A persist 0.03 → 0.16, A→BB 0.29 → 0.21, §4). It is *not* a non-identifiability. The type composition, the early endpoints and the higher moments carry further constraints. A naive compensating move that matches M and $\operatorname{Var} W$ exactly (A: div_AA and div_BB −0.22, div_AB +0.44) is *easier* to detect than the uncompensated B move: KL 0.17–0.28 vs 0.011–0.016 nats per clone. The cheapest compensations are found numerically in §5. They raise A *death*, not A persist: B→AA instead of B→A makes extra A cells, and A death removes them. The NPE answer at 4096 (A persist 0.15) is a suboptimal version of the same compensation.

Background and references. The approximation $Z_n \approx \rho^n W u$ (Kesten–Stigum): [B5], [B1, Ch. V]; the Gaussian fluctuations of the type composition around u: [B8], [B1, Ch. V]. Parameters compensating each other along a valley ("sloppiness"): [I4].

3 Exact likelihood: local identifiability and its strength

The law of Z_n is computed exactly by evaluating F_n^A on an $L \times L$ grid of roots of unity and applying a 2-D FFT. The polynomial F_n^A has degree $\leq 2^n$ in each variable, so $L = 2^n + 1$ is

alias-free. For $n = 10$, $L = 384$ already reproduces the alias-free result to 10^{-17}, because the mass beyond 384 cells is below 10^{-14}. Derivatives $\partial_\theta F_n$ are propagated by forward-mode differentiation through the iteration $F_n = f(F_{n-1})$ and checked against finite differences (relative error 10^{-11}). Means agree with $e_A M^n$ to machine precision. The Fisher information of the *conditioned* law $\tilde{p} = p/(1 - p_{00})$ is $I_n = \sum_{z \neq 0} \tilde{p}_z \nabla \log \tilde{p}_z \nabla \log \tilde{p}_z^\top$, in the 10 coordinates "entry minus death" of each row. Because $\theta^\star$ is on the boundary, B-division directions are one-sided. A positive definite I_n still gives $\mathrm{KL}(\theta^\star \| \theta) \geq c\,|\theta - \theta^\star|^2$ on the admissible cone near $\theta^\star$. Code: `scripts/b_row_theory/`.

Computation 6 (Local identifiability at $\theta^\star$)**.** For every single endpoint $n \in \{2, \ldots, 10, 12\}$ the per-clone Fisher information $I_n(\theta^\star)$ of the 10-parameter model is positive definite (Table 1). Hence $\theta^\star$ is locally identifiable from *any single* endpoint distribution. The smallest eigenvalue is far above floating-point error: its ratio to the largest is $\geq 5 \times 10^{-7}$, against a relative accuracy of $\approx 10^{-12}$ for the computed probabilities.

Background and references. Inverting a pgf by evaluating it on the unit circle and applying an FFT: [N1]; forward-mode differentiation through an iteration: [N2]. Fisher information and its link to local identifiability: [I1], [I5, Ch. 2]. Score of a conditioned (truncated) distribution: derived from $\log \tilde{p} = \log p - \log(1 - p_{00})$; any treatment of truncated likelihoods, e.g. [I9]. Behaviour at a boundary point ($\delta = 0$): [I7]. Fisher spectra spanning many decades (Table 1): [I4].

n	$\lambda_{\min}(I_n)$	cond. I_n	CR var. B→A	CR var. δ	N for sd 0.05
2	0.362	1.3×10^2	1.8	1.7	730
3	0.331	1.9×10^2	2.3	1.8	920
4	0.205	5.1×10^2	3.6	3.6	1 440
5	0.051	3.2×10^3	13	14	5 300
6	0.037	6.6×10^3	18	19	7 200
7	0.015	2.3×10^4	41	39	16 000
8	0.0090	5.3×10^4	64	54	26 000
9	0.0044	1.4×10^5	124	91	50 000
10	0.0023	3.5×10^5	218	143	87 000
12	0.00067	1.8×10^6	686	413	274 000

Table 1: Exact per-clone Fisher information at $\theta^\star$, single endpoint, 10 free parameters. CR variance of B→A and of the B-division mass $\delta = a_B + b_B + c_B$ with all other parameters free (profiled), per clone: divide by N. Last column: N for a CR sd of 0.05 on B→A from that endpoint alone.

The weak direction. The eigenvector of the smallest eigenvalue of $I_{\mathcal{D}}$ is, at every design checked, dominated by B→A (≈ 0.8) against B→AA and B→AB (-0.2 to -0.5), with a smaller B persist component. This is the switching-versus-division direction of Theorem 2 (move e_1, partly e_3), now as a property of the full likelihood rather than of the means.

Proposition 7 (Practical non-identifiability at realistic designs)**.** *CR bounds for B→A and for δ (all other parameters free) are listed in Table 2. At the real-like and E0 designs the bound on B→A is 0.27–0.28. The uniform Dirichlet prior has sd 0.14 (Beta(1,5)) and the flat prior on B→A has sd 0.29. So no estimator that is unbiased near $\theta^\star$ can be more precise than the prior, and the posterior is necessarily prior-dominated. This is the regime of E0, X5a and X5c, and their conclusions hold there.*

Background and references. Cramér–Rao bound, scalar and for a function $g(\theta)$ with nuisance parameters, $g^\top I^{-1} g$: [I5, Ch. 2], [I6, Ch. 8]. Fisher information adds over independent observations, so $I_{\mathcal{D}} = \sum_n N_n I_n$: [I5, Ch. 2]. Practical identifiability: [I2, I3]. Prior vs data at small samples, and why a posterior can beat the CR bound only by being biased: [I6, Ch. 10], [S10]. Comparing designs by the CR variance of one parameter is c-optimal design: [D1], [D2].

design (endpoint: clones)	organoids	CR sd B→A	CR sd δ
real-like $\{6{:}165, 10{:}101\}$	266	0.28	0.28
E0 $\{4, 8, 12\} \times 32$	96	0.27	0.28
$\{4, 6, 10\} \times 512$	1 536	0.065	0.068
E0 at 4096: $\{4, 6, 10\} \times 4096$	12 288	0.023	0.024
$\{4\} \times 266$	266	0.12	0.12
$\{3\} \times 266$	266	0.093	0.083
$\{2, 3\} \times 133$	266	0.077	0.066
$\{3, 6, 10\} \times \{89, 89, 88\}$	266	0.12	0.12
$\{2, 4, 6, 10\} \times \approx 66$	266	0.093	0.091
$\{3, 4\} \times 1000$	2 000	0.036	0.033

Table 2: Cramér–Rao sd at $\theta^\star$ (exact Fisher information, GW, generations). Top: designs used in the report. Bottom: same 266 organoids redistributed to early endpoints.

Proposition 8 (The information about B's row is in early generations)**.** *Per clone, the information about B→A and δ decreases geometrically with the endpoint. From $n = 5$ to $n = 12$ the CR variance of B→A grows by a factor 1.4–2.3 per generation (geometric mean 1.76; Table 1), and the condition number of I_n by a factor 2–3.5. Between $n = 2$ and $n = 4$ the loss is milder (factor 2 in total).*

Interpretation. At small n the counts are small integers, and individual events are visible in the law. A clone $(2, 0)$ at $n = 2$ arises from A→B→AA but not from A→B→A. As n grows the law of Z_n approaches $\rho^n W u$ plus Gaussian composition noise. Its first moments are blind to the fibres of Theorem 2, and its main second moment ($\operatorname{Var} W$) mixes the two rows (Theorem 4). The geometric rate is a numerical observation over $n \leq 12$, not a proved asymptotic.
Consequence for design — local versus global. Locally, with the same 266 organoids, an endpoint at generation 2–3 gives a CR sd of 0.08–0.09 on B→A, against 0.28 at $\{6, 10\}$: 10–15 times the Fisher information. Adding one early endpoint to $\{6, 10\}$ already helps (0.12 for $\{3, 6, 10\}$). *Globally* the gain is smaller. The early-endpoint likelihood is quadratic only up to $\delta \approx 0.1$ and then flattens (Table 3): the best "B divides half the time" model (B→A ≈ 0.11) costs only 2.7 nats at $\{3\} \times 266$ and 6.7 nats at $\{2, 3\} \times 133$, against 18 and 29 nats predicted by the Fisher information. That is still 7–18 times the evidence of the real-like design (0.36 nats), but comparable to the uniform prior's penalty on B→A $= 0.8$ (≈ 6 nats). Exact posterior sampling confirms it (§4, 5 seeds): B→A $= 0.23$ (sd 0.19, 0/5 covered) at $\{3\} \times 266$, and 0.56 (sd 0.15, 1/5 covered, B-division mass 0.18) at $\{2, 3\} \times 133$ (truth 0.8), against 0.25 at the real-like design. Ten nats of evidence against $\delta = 0.5$ need $\approx 1\,000$ organoids at $\{3\}$, ≈ 400 at $\{2, 3\}$ and $\approx 7\,400$ at $\{6, 10\}$. So early endpoints are the most efficient design, but at realistic sizes they do not by themselves make B's row identifiable; combined with a biological constraint or a measurement of B divisions they help most. Whether generation 2–3 is reachable in the wet lab (day 2–3 for a ≈20 h A cycle) is open (§7).

δ	0.05	0.1	0.2	0.3	0.5	0.7
$\{3\} \times 266$: $266\,K(\delta)$ [nats]	0.16	0.56	1.6	2.5	2.7	3.2
Fisher prediction	0.18	0.72	2.9	6.5	18	35
$\{2, 3\} \times 133$: $266\,K(\delta)$ [nats]	0.28	1.0	3.4	5.6	6.7	10.6
Fisher prediction	0.29	1.2	4.6	10	29	57
real-like $\{6{:}165, 10{:}101\}$ [nats]	0.014	0.05	0.15	0.25	0.36	—

Table 3: Global profile (expected log-LR of the truth against the best model with B-division mass δ, 266 organoids) for early-endpoint designs, against the local Fisher prediction $266\,\delta^2/(2\sigma_\delta^2)$. Code: `scripts/b_row_theory/profile_early.py`.

Background and references. Long-time behaviour of the clone ($\rho^n W u$ and composition fluctuations): [B5], [B8], [B1, Ch. V]. The biological counterpart, long-term clone-size distributions converging to universal

scaling forms that forget fate details: [C3], [C4]; using short-term clonal data to resolve fate choices: [C1], [C2]. Choosing measurement times from the Fisher information: [D1], [D2].

4 E0 at 4096 clones revisited

The report (§ "E0 revisited at 4096 clones") concludes "structural non-identifiability" because, at $\{4, 6, 10\} \times 4096$ with a 20k-simulation NPE, B→A stays at 0.17–0.34 (truth 0.8) and the B-division mass at its prior mean. Three exact checks contradict the "structural" reading for GW.

1. The bound. The CR sd is 0.023 (Table 2). NPE reported a posterior sd of 0.11 and a bias of 0.47, which is 20 CR sd.

2. The exact likelihood rejects the NPE answer on the report's own data. For each run we take the NPE posterior mean $\hat{\theta}$, renormalise each row, and evaluate the exact conditioned log-likelihood on the stored reference data (`reference.npz`, 3×4096 clones):

run (seed)	0	1	2	3	4
20k: $\ell(\theta^\star) - \ell(\hat{\theta})$ [nats]	149.8	137.7	122.4	151.7	149.6
5k: $\ell(\theta^\star) - \ell(\hat{\theta})$ [nats]	168.9	164.3	116.5	177.6	170.3
expected, $4096 \sum_n \mathrm{KL}_n(\theta^\star \| \hat{\theta})$ (20k)	124	141	130	132	139

A likelihood ratio of e^{116}–e^{178} is decisive. The NPE posterior concentrates on parameters that the data exclude. The clean PPC does not contradict this: PPC compares *summaries* within 95% bands, a far coarser check than the likelihood.

3. The summaries carry the information; the network does not use it. We simulated 300 datasets at $\theta^\star$ and 300 at $\hat{\theta}$ (seed 0, 20k) with the `moments` summary. A regularised logistic regression separates them with 5-fold CV accuracy **0.968** (0.722 at 165 clones per endpoint). The most shifted features belong to the earliest endpoint (h4: sd of $\log(1+B)$, median of $\log(1+A)$) and to the A–B correlation. In Gaussian-equivalent terms, accuracy 0.968 corresponds to an expected log-LR of ≈ 7 nats. So the summary keeps about 5% of the exact 124 nats along this direction, and still more than enough to reject $\hat{\theta}$ ($\approx 900{:}1$). The amortised NPE, trained on 20k draws spread over a 10-dimensional prior, does not resolve this.

4. Exact posterior sampling confirms it. The `exact_gw` inference method (branch `exact-gw`: NUTS on the exact likelihood, same prior and coordinates as NPE) was run on the *same* reference datasets as the NPE runs (5 seeds, data seed = seed). Results are in `results/exact_gw/`; table `exact_vs_npe.csv` from `scripts/analyse_exact_gw.py`:

design	law	method	B→A (0.8)	post. sd	covered	B-div. (0)
E0 at 4096, $\{4, 6, 10\}$	`two_type`	exact	0.778	0.020	4/5	0.04
		NPE 20k / 5k	0.34 / 0.27	0.12 / 0.14	0/5	0.49 / 0.56
E0, $\{4, 8, 12\} \times 32$	`two_type`	exact	0.27	0.17	0/5	0.52
		NPE 5k–50k	0.20–0.22	0.15	0/5	0.54
E0, $\{4, 8, 12\} \times 32$	`reversible_ab`	exact	0.77	0.10	5/5	—
		NPE 5k / 20k	0.72 / 0.71	0.13	5/5	—
real-like	`two_type`	exact	0.25	0.16	0/5	0.56
real-like	`reversible_ab`	exact	0.77	0.08	5/5	—
		NPE 5k (10 seeds)	0.75	0.12	10/10	—
$\{3\} \times 266$	`two_type`	exact	0.23	0.19	0/5	0.54
$\{2, 3\} \times 133$	`two_type`	exact	0.56	0.15	1/5	0.18

Means over seeds; covered = 90% interval contains the truth. Caveats: no SBC for the `two_type` E0 runs; two runs have $\hat{R}$ 1.12 and 1.18 (multimodal posteriors), all others ≤ 1.11. At 4096 clones exact inference recovers B→A with a posterior sd (0.020) at the Cramér–Rao scale (0.023) and puts almost no mass on B divisions, where NPE was off by 0.44–0.53: the failure was NPE's. Where the data are weak (E0 design, real-like with B divisions free) exact inference agrees with NPE, so NPE's answers there were the data's. The early-endpoint rows show that a local bound can be misleading (Table 3).

Conclusion. In GW the B row is not structurally non-identifiable. It is *practically* non-identifiable at the designs of E0, X5a, X5c and the real-like design (Proposition 7). At 4096 clones per endpoint it is identifiable from the data, from the exact likelihood and to a useful extent from the summaries, and the failure reported there is an inference failure (amortised NPE with uninformative coverage of the relevant region). The report's recommendations survive: fix or measure B divisions, because realistic designs cannot decide them. Their justification and wording should change (§6).

Background and references. Expected log-likelihood ratio = $N\times$KL, and KL as the rate of hypothesis testing (Stein's lemma): [I8, Ch. 11]. Unfaithful (over-confident or biased) NPE posteriors: [S6]; amortised vs sequential or truncated training: [S2, S4]. Why posterior predictive checks on summaries are weak: [S10, Ch. 6]. Classifier two-sample tests: [S11]. Converting a classifier accuracy into a Gaussian-equivalent separation d' (accuracy $= \Phi(d'/2)$) and KL $= d'^2/2$ for equal-variance Gaussians: [S14], [I8, Ch. 8]. Information lost by summary statistics (sufficiency, and how to construct better summaries): [S7], [S8].

5 Profile: how flat is the likelihood away from $\theta^\star$?

The Fisher information is local. For the global picture we compute the profile

$$K_{\mathcal{D}}(\delta) = \min\Big\{ \sum_n w_n \,\mathrm{KL}\left(\tilde{p}_n(\theta^\star) \,\|\, \tilde{p}_n(\theta)\right) :\ \theta \in \Delta_5 \times \Delta_5,\ a_B + b_B + c_B = \delta \Big\},$$

the per-clone cost of the best-fitting model in which B divides with probability δ. All other 9 coordinates are optimised (L-BFGS, three starts, exact law). Then $N\,K(\delta)$ is the expected log-likelihood ratio against the best "B divides" explanation with N organoids in total, split in the design's proportions ($w_n = N_n/N$).

B-division mass δ	0.02	0.05	0.1	0.2	0.3	0.5
real-like {6:165, 10:101} (weights 165/266, 101/266)						
$K(\delta)$ per clone	$9.8\cdot10^{-6}$	$5.4\cdot10^{-5}$	$1.9\cdot10^{-4}$	$5.8\cdot10^{-4}$	$9.3\cdot10^{-4}$	$1.36\cdot10^{-3}$
$266\,K(\delta)$ [nats]	0.003	0.014	0.05	0.15	0.25	0.36
best B→A / A death	0.78 / 0.04	0.76 / 0.05	0.71 / 0.07	0.62 / 0.10	0.55 / 0.12	0.39 / 0.16
E0 at 4096 {4, 6, 10}, equal weights						
$K(\delta)$ per clone	$2.9\cdot10^{-5}$	$1.6\cdot10^{-4}$	$5.5\cdot10^{-4}$	$1.6\cdot10^{-3}$	$2.8\cdot10^{-3}$	$4.2\cdot10^{-3}$
$12\,288\,K(\delta)$ [nats]	0.35	1.9	6.7	20	34	51
best B→A / A death	0.78 / 0.04	0.76 / 0.05	0.72 / 0.07	0.64 / 0.10	0.54 / 0.12	0.33 / 0.15

Table 4: Profile KL $K(\delta)$ (truth $\delta = 0$, B→A = 0.8, A death 0.03) and the B→A and A death of the best-fitting model at each δ. Expected log-LR of truth vs. that model = number of organoids $\times K(\delta)$. Full minimisers: `scripts/b_row_theory/profile.jsonl`.

Computation 9 (Global profile)**.** (i) $K(\delta) > 0$ for every δ on the grid. Near 0 it matches the Fisher prediction $\delta^2/(2\sigma_\delta^2)$, with σ_δ^2 the per-clone CR variance of the design (E0 at 4096: predicted $2.8\cdot10^{-5}$ at $\delta = 0.02$, computed $2.9\cdot10^{-5}$). Above $\delta \approx 0.2$, K grows sub-quadratically: explanations in which B divides a lot are relatively cheap. (ii) *Real-like design:* the best model with $\delta = 0.5$ (B→A = 0.39) has an expected log-LR of only 0.36 nats against the truth, a likelihood ratio of

about 1.4. Every $\delta \leq 0.5$ is statistically indistinguishable from $\delta = 0$ with 266 organoids. This is the non-local form of Proposition 7. Three nats of expected evidence against $\delta = 0.5$ would need $\approx 2\,200$ organoids in the real-like split, or ≈ 720 split equally over $\{4, 6, 10\}$. (iii) *E0 at 4096:* the same alternative costs 51 nats, so it is excluded. Its parameters (B→A 0.33, A death 0.15, B→AA 0.34) are close to the NPE posterior mean (B→A 0.32, A death 0.09, B→AA 0.21). NPE found the bottom of the valley but not its depth. (iv) *The compensation* is mainly move e_1 (switch → divide-into-AA) in B's row, with the extra death *moved to A's row.* A death rises from 0.03 to 0.16 while B death stays at 0.06–0.10. This is the precise form of "the confounding leaks into A's row".

X5c ($\{6{,}10\}\times\{32{,}128{,}512\}$) is consistent with this. Using the real-like profile as an approximation (its split differs slightly), 1 024 organoids give ≈ 1.4 nats of expected evidence against $\delta = 0.5$. So its failure is mainly the likelihood's, not only the network's.

Background and references. Profile likelihood: [I2], [I9]. Minimising $\mathrm{KL}(\theta^\star \| \theta)$ over a restricted model gives the "pseudo-true" parameter that the MLE of that model converges to: [I10], [I6, Ch. 5]. The small-δ approximation $K(\delta) \approx \delta^2/(2\sigma_\delta^2)$ is the second-order expansion of KL, whose Hessian is the Fisher information: [I8, Ch. 11], [I5, Ch. 2]. L-BFGS-B: [N3, Ch. 7].

6 Suggested changes to the report's wording

- Replace "structural non-identifiability" (E0 at 4096, Part I summary, "Decided: structural") with: *"practically non-identifiable at realistic designs (Cramér–Rao sd of B→A 0.27–0.28 > prior sd); locally identifiable in principle (positive definite exact Fisher information); at 4096 clones per endpoint the data and the summaries identify it, but amortised NPE does not."*
- "A clean PPC is no evidence that the B row is right" stays true. Add: the exact likelihood ratio against the NPE answer is e^{120}–e^{180} on the same data.
- "Biological constraints help more than data" holds at realistic sizes, with the caveat that early data help several times more than late data per organoid (Proposition 8, Table 3), although at 266 organoids not enough to replace the constraint.
- E0's first attribution (prior tail) and its revision stay as they are. The revision "the confounding is the root cause" is correct as a description of practical non-identifiability.

7 Scope, caveats and open questions

- **Global identifiability** (injectivity) is not proved. The profile of §5 is numerical evidence over a grid of δ, from finitely many starts.
- **One truth.** All computations are at $\theta^\star$. Theorems 2–4 hold for every θ; the Fisher numbers do not. The same scripts compute them at any other truth (e.g. the X5a truths).
- **Boundary.** $\theta^\star$ has $\delta = 0$ on the boundary. The MLE of δ is asymptotically a censored normal, and the CR bound is the local scale, not an exact sd.
- **CR vs posterior.** The CR bound concerns unbiased estimators. Posteriors with informative priors can have smaller sd but are then biased, which is the E0/X5 picture. Asymptotically (data dominating) the posterior sd approaches the CR sd.
- **Bellman–Harris (local analysis only).** An exact BH likelihood for integer Gamma shapes (phase ODEs + FFT; `bhexact.py`, branch `bh-exact`, method in `exact_likelihood.pdf`) was validated against the simulator. At the real-like BH design $\{144\,\text{h}{:}165, 240\,\text{h}{:}101\}$ its Fisher information is non-singular, so the BH ridge (Part III) and the B-row problem are

practical, not structural. They are much weaker than in GW. CR sd of B→A: 0.82 with B divisions fixed and 0.99 with them free (GW: 0.28 free). CR sd of A→AA: 0.23 fixed, 0.54 free. The weakest direction trades B→A against B persist and B's mean lifetime. In BH, fixing B divisions is therefore not enough; B's lifetime also has to be known. Profiles and exact posteriors for BH are not done, and the early-endpoint lesson above suggests the local numbers may again overstate what the data can do.

- **Generations vs days.** The early-endpoint recommendation is in GW generations. Its translation to days, and to what can be counted and typed in 2–8-cell organoids, is a wet-lab question.
- **Inference fix at 4096.** The analysis suggests likelihood-based inference (exact GW likelihood here, or sequential/local NPE around the data), or summaries targeted at early endpoints. It does not show that a particular NPE configuration would succeed.

8 A toolkit for similar situations

The analysis above follows a sequence that applies to any simulator-based model where a parameter "is not recovered". Each step names the readings in §9 that teach it.

1. **Look for exact invariances first** (moments, symmetries). Which parameter combinations leave the mean, the variance, etc. unchanged? (Theorems 2–4; [B1, B2, B5].)
2. **Separate structural from practical.** Is the Fisher information singular (structural, at least locally), or just small? A singular information means a flat likelihood; a small one means more or better data. ([I1–I4].)
3. **Profile the likelihood** along the suspect direction, re-optimising everything else. It shows how deep the valley is, globally and not only near the truth, and which other parameters compensate. ([I2, I3].)
4. **Compare the method with the bound.** If the posterior is much wider or more biased than the Cramér–Rao bound, the method is losing information. Check where: exact likelihood ratio on the observed data, then a classifier two-sample test on the summaries, then the network. ([S5–S8, S11].)
5. **Use the information to design.** Fisher information per organoid and per endpoint says which measurements help; this is (local) optimal experimental design. ([D1, D2].)
6. **Treat clean diagnostics with care.** SBC averages over the prior, and a PPC on summaries is coarser than the likelihood. Neither shows identifiability. ([S6, S9, S10].)

9 Further reading

Suggested order for a first pass: B3 or B4 (branching processes for biologists), I2 and I3 (identifiability in practice), S1 and S8 (simulation-based inference and its failure modes), then D2. Starred items are the most relevant to this note. Each result above ends with a "Background and references" paragraph pointing here. Chapter numbers refer to the editions listed. They were given from memory, so check the table of contents if a pointer seems off.

Branching processes (the model)

B1⋆ K. B. Athreya and P. E. Ney, *Branching Processes*, Springer, 1972 (Dover reprint 2004). Ch. V: multitype processes, the mean matrix, moment recursions and the Kesten–Stigum limit, i.e. the background of Theorems 3 and 4.

B2 T. E. Harris, *The Theory of Branching Processes*, Springer, 1963. The classical source for probability generating functions and the covariance recursion (1); also covers age-dependent (Bellman–Harris) processes.

B3⋆ M. Kimmel and D. E. Axelrod, *Branching Processes in Biology*, 2nd ed., Springer, 2015. The most accessible entry point; cell proliferation, multitype and age-dependent models, with biological examples.

B4 P. Haccou, P. Jagers and V. A. Vatutin, *Branching Processes: Variation, Growth, and Extinction of Populations*, Cambridge University Press, 2005. Population-biology viewpoint; good on multitype growth and extinction.

B5 H. Kesten and B. P. Stigum, "A limit theorem for multidimensional Galton–Watson processes", *Ann. Math. Statist.* 37 (1966) 1211–1223. Origin of $Z_n \approx \rho^n W u$.

B6 P. Guttorp, *Statistical Inference for Branching Processes*, Wiley, 1991. Likelihood and estimation for branching processes, including what can and cannot be estimated from generation counts.

B7⋆ W. Feller, *An Introduction to Probability Theory and Its Applications*, Vol. I, 3rd ed., Wiley, 1968. Ch. XI: generating functions, moments from derivatives, and the branching-process composition rule, at the most elementary level.

B8 H. Kesten and B. P. Stigum, "Additional limit theorems for indecomposable multidimensional Galton–Watson processes", *Ann. Math. Statist.* 37 (1966) 1463–1481. Central limit theorems for the type composition around the Perron direction.

Clonal dynamics in stem-cell biology (the application)

C1⋆ H. J. Snippert et al., "Intestinal crypt homeostasis results from neutral competition between symmetrically dividing Lgr5 stem cells", *Cell* 143 (2010) 134–144. LGR5$^+$ clone-size data interpreted with stochastic fate models.

C2 E. Clayton et al., "A single type of progenitor cell maintains normal epidermis", *Nature* 446 (2007) 185–189. A classic inference of division and fate probabilities from clone-size distributions.

C3 A. M. Klein and B. D. Simons, "Universal patterns of stem cell fate in cycling adult tissues", *Development* 138 (2011) 3103–3111. Why clone-size distributions often fix only some combinations of fate parameters.

C4 S. Rulands and B. D. Simons, "Tracing cellular dynamics in tissue development, maintenance and disease", *Curr. Opin. Cell Biol.* 43 (2016) 38–45. Short review of lineage tracing and its quantitative analysis.

Identifiability, Fisher information, profile likelihood

I1 T. J. Rothenberg, "Identification in parametric models", *Econometrica* 39 (1971) 577–591. Local identifiability ⇔ non-singular Fisher information (under regularity), the basis of Computation 6.

I2⋆ A. Raue et al., "Structural and practical identifiability analysis of partially observed dynamical models by exploiting the profile likelihood", *Bioinformatics* 25 (2009) 1923–1929. The structural vs practical distinction and the profile-likelihood method used in §5.

I3⋆ F.-G. Wieland, A. L. Hauber, M. Rosenblatt, C. Tönsing and J. Timmer, "On structural and practical identifiability", *Curr. Opin. Syst. Biol.* 25 (2021) 60–69. A recent, readable review of the same distinction, with pitfalls.

I4 R. N. Gutenkunst et al., "Universally sloppy parameter sensitivities in systems biology models", *PLoS Comput. Biol.* 3 (2007) e189; and M. K. Transtrum et al., "Perspective: Sloppiness and emergent theories in physics, biology, and beyond", *J. Chem. Phys.* 143 (2015) 010901. Why Fisher eigenvalues spread over many decades (cf. Table 1) and what that means.

I5 E. L. Lehmann and G. Casella, *Theory of Point Estimation*, 2nd ed., Springer, 1998. Fisher information and the Cramér–Rao bound, with proofs.

I6 A. W. van der Vaart, *Asymptotic Statistics*, Cambridge University Press, 1998. Asymptotics of the MLE and of posteriors (Bernstein–von Mises): why the CR bound is the right yardstick for large samples.

I7 S. G. Self and K.-Y. Liang, "Asymptotic properties of maximum likelihood estimators and likelihood ratio tests under nonstandard conditions", *J. Amer. Statist. Assoc.* 82 (1987) 605–610. Parameters on the boundary (here $\delta = 0$).

I8⋆ T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd ed., Wiley, 2006. Ch. 11 (information theory and statistics): KL divergence as the expected log-likelihood ratio, Stein's lemma, and the link between KL and Fisher information. Ch. 8: relative entropy of continuous (e.g. Gaussian) distributions.

I9 Y. Pawitan, *In All Likelihood: Statistical Modelling and Inference Using Likelihood*, Oxford University Press, 2001. Likelihood-based inference for applied scientists: Fisher information, profile likelihood, likelihood ratios. A good textbook companion to [I2].

I10 H. White, "Maximum likelihood estimation of misspecified models", *Econometrica* 50 (1982) 1–25. The KL-closest ("pseudo-true") parameter of a restricted or wrong model, as computed in §5.

Simulation-based inference: methods, summaries, diagnostics

S1⋆ K. Cranmer, J. Brehmer and G. Louppe, "The frontier of simulation-based inference", *PNAS* 117 (2020) 30055–30062. Overview of NPE, NLE, NRE and where information is lost.

S2 G. Papamakarios and I. Murray, "Fast ϵ-free inference of simulation models with Bayesian conditional density estimation", NeurIPS 2016; D. S. Greenberg, M. Nonnenmacher and J. H. Macke, "Automatic posterior transformation for likelihood-free inference", ICML 2019. Neural posterior estimation, amortised and sequential.

S3 G. Papamakarios, D. Sterratt and I. Murray, "Sequential neural likelihood", AISTATS 2019. Learning the likelihood instead of the posterior; an alternative when the posterior is hard to amortise.

S4 M. Deistler, P. J. Gonçalves and J. H. Macke, "Truncated proposals for scalable and hassle-free simulation-based inference", NeurIPS 2022. Concentrating simulations near the data: the remedy suggested by §4.

S5 J.-M. Lueckmann et al., "Benchmarking simulation-based inference", AISTATS 2021. How methods compare against reference posteriors, the logic of item 4 of §8.

S6⋆ J. Hermans et al., "A trust crisis in simulation-based inference? Your posterior approximations can be unfaithful", *TMLR* 2022. Over-confident or biased NPE posteriors, as at 4096 clones.

S7 P. Fearnhead and D. Prangle, "Constructing summary statistics for approximate Bayesian computation: semi-automatic approximate Bayesian computation", *J. R. Stat. Soc. B* 74 (2012) 419–474; Y. Chen et al., "Neural approximate sufficient statistics for implicit models",

ICLR 2021. Choosing or learning summaries that keep the information (cf. the 5% kept by `moments`).

S8⋆ S. A. Sisson, Y. Fan and M. Beaumont (eds.), *Handbook of Approximate Bayesian Computation*, CRC Press, 2018. Broad reference on likelihood-free inference, summary statistics and diagnostics.

S9 S. Talts, M. Betancourt, D. Simpson, A. Vehtari and A. Gelman, "Validating Bayesian inference algorithms with simulation-based calibration", arXiv:1804.06788 (2018). SBC and what it does *not* check.

S10 A. Gelman et al., *Bayesian Data Analysis*, 3rd ed., CRC Press, 2013, Ch. 6. Posterior predictive checks and their limits.

S11 D. Lopez-Paz and M. Oquab, "Revisiting classifier two-sample tests", ICLR 2017. The C2ST used in E0 and the two-point test of §4.

S12 S. N. Wood, "Statistical inference for noisy nonlinear ecological systems", *Nature* 466 (2010) 1102–1104. Synthetic likelihood: a Gaussian model of the summaries. Note its bias with noisy finite-difference derivatives, which made us drop such an estimate here.

S13 A. Tejero-Cantero et al., "sbi: A toolkit for simulation-based inference", *J. Open Source Softw.* 5 (2020) 2505. The library used by this project.

S14 D. M. Green and J. A. Swets, *Signal Detection Theory and Psychophysics*, Wiley, 1966. The discriminability d' and its relation to classification accuracy for two Gaussian classes.

Probability and numerical methods

P1 D. Williams, *Probability with Martingales*, Cambridge University Press, 1991. Ch. 12: martingales bounded in L^2 and orthogonality of increments, used in Theorem 4. Also a short treatment of the Galton–Watson martingale.

N1 J. Abate and W. Whitt, "Numerical inversion of probability generating functions", *Oper. Res. Lett.* 12 (1992) 245–251. Recovering probabilities from a pgf on a circle by discrete Fourier transform, with aliasing control: the method of `gwexact.py`.

N2 A. Griewank and A. Walther, *Evaluating Derivatives: Principles and Techniques of Algorithmic Differentiation*, 2nd ed., SIAM, 2008. Forward-mode (tangent) differentiation, as used for $\partial_\theta F_n$.

N3 J. Nocedal and S. J. Wright, *Numerical Optimization*, 2nd ed., Springer, 2006. Ch. 7: limited-memory BFGS, the optimiser of the profile.

Experimental design

D1 A. C. Atkinson, A. N. Donev and R. D. Tobias, *Optimum Experimental Designs, with SAS*, Oxford University Press, 2007. Fisher-information-based (D-, A-, c-optimal) design; the c-optimal criterion for one parameter is exactly the CR variance of Table 2.

D2⋆ K. Chaloner and I. Verdinelli, "Bayesian experimental design: a review", *Statist. Sci.* 10 (1995) 273–304. Design when the prior matters, as it does at realistic sizes here.

Reproducibility. `scripts/b_row_theory/`: `gwexact.py` (exact law, derivatives, Fisher information, KL), `fisher.py`, `fisher2.py` (Tables 1, 2), `llr.py` (§4, item 2; reads `results/e4/gw_xxl_two_type_*`), `summary_twopoint.py` (item 3), `varW.py` (Theorem 4 check), `profile.py` (§5). Run with `uv run python scripts/b_row_theory/<script>.py` from the repository root.